

GreenGuard Module 1 — Leaf Disease Model Evaluation

EfficientNet-B0 (efficientnet_b0-1.0), 40 classes, 7 crops · 528 held-out photographs · 26 September 2026

1. Verdict

Usable for three crops. Not usable for rice or wheat.

Given the crop, the model identifies the disease correctly **56% of the time** on its first guess, against 19% for random guessing. Within its three-item shortlist the correct disease appears **83% of the time**. Accuracy is highly uneven across crops: potato, cotton and maize reach 73–77% first-guess accuracy, while rice and wheat reach only 34% — close to guesswork for those two crops. No threshold setting corrects this; it requires retraining.

The service is therefore configured to present a **ranked shortlist of three** rather than a single diagnosis. The shortlist is the product; the single best guess is not reliable enough to stand alone.

2. What was tested, and how

528 photographs covering all 33 disease classes across the 7 supported crops, sourced from the web rather than from the training distribution. Each image's true class was read from its folder path and filename code (for example wheat/yellow_rust/Wyr3.jpg); all 528 resolved unambiguously and none were excluded.

Every measurement supplies the model with the **correct crop**, because in the application the farmer selects the crop before photographing. Identifying the crop is therefore never the model's task, and figures here reflect only its ability to tell that crop's diseases apart.

Because the photographs are web-sourced, this measures **generalisation to realistic field imagery**, which is a harder and more relevant test than a held-out split of the training data. A higher accuracy reported during training is not contradicted by these figures; the gap between them is itself the finding.

3. Overall accuracy, crop supplied

Measure	Result	Random guessing	Reading
First guess correct (top-1)	56%	19%	roughly 3× chance
Correct within top 2	73%	—	
Correct within top 3	83%	—	the shortlist usually contains it

Measured with all decision gates removed, so these reflect the model itself rather than the thresholds applied on top of it.

4. Accuracy by crop

The single most important table in this report. A shortlist of three drawn from a crop with only three classes conveys nothing, so the shortlist figure must always be read against what pure chance would already achieve.

Crop	Classes	Top-1	Top-3	Top-3 by chance	Verdict
Potato	3	77%	100%	100%	usable
Cotton	4	73%	93%	75%	usable
Maize	4	73%	96%	75%	usable
Sugarcane	5	61%	92%	60%	borderline
Tomato	9	48%	80%	33%	weak
Rice	6	34%	62%	50%	do not ship
Wheat	9	34%	59%	33%	do not ship

Two entries need reading carefully

Potato — 100% top-3 is arithmetic, not skill. Only three potato classes exist, so a three-item shortlist lists all of them. For potato the interface should show the single prediction (77% correct) and no shortlist.

Rice — 62% top-3 against 50% by chance. Neither the first guess (34%) nor the shortlist carries meaningful signal. Rice should not be offered to farmers until the model is retrained.

5. The confidence threshold and what it buys

The model returns a confidence score, and the service stays silent when that score falls below a configured threshold. The threshold therefore governs **how often the service answers**, not how often it is right. Lowering it does not make the model better; it makes it speak more, correctly and incorrectly alike.

Threshold	Answers given	First guess right	Shortlist has it	Wrong first guesses
0.00	100%	56%	83%	231
0.25	95%	59%	85%	206
0.35 (in use)	86%	63%	88%	165
0.50	74%	67%	89%	129
0.70	56%	72%	92%	83
0.95	25%	81%	93%	25

0.35 was chosen so the service answers 86% of photographs rather than 25%. The cost is that its first guess is then wrong on roughly one answer in three — acceptable only because the shortlist is shown alongside it, and contains the correct disease 88% of the time.

Requirement on the application, not a suggestion

The interface must present all three ranked possibilities, never the single prediction as a diagnosis. Displaying only the first guess would place an incorrect disease in front of a farmer in roughly one case in three, with a treatment decision attached to it. Suggested phrasing: “most likely X; possibly Y or Z — confirm before treating”.

6. Configuration applied

All values live in `model/config.json` and are changed without touching code.

Setting	Was	Now	Why
crop_filtered_threshold	0.80	0.35	answer 86% of photographs instead of 45%, with the shortlist carrying the accuracy
crop_mismatch_threshold	0.50	1.01	disabled: the farmer's crop selection is authoritative, and the model's own crop guess was wrong on 38% of photographs, so the check rejected good answers
confidence_threshold	0.70	0.70	unchanged; applies only if no crop is supplied, which the application never does

7. Limits of this evaluation

Limitation	Consequence
No healthy leaves were tested. The set contains 33 disease classes and no healthy class, though the model has one for every crop.	The rate at which the model reports disease on a <i>healthy</i> plant is unmeasured. For a farmer-facing tool this is the most damaging error available, and it is currently unknown. Approximately 50 healthy photographs per crop should be tested before release.
One test set, web-sourced.	Figures describe generalisation to web imagery. Photographs taken by farmers on low-end phones may differ again.
Class sizes are uneven (10 to 25 photographs per class).	Per-crop figures drawn from 10 photographs carry a margin of roughly ± 15 points; small differences between crops should not be over-read.
Non-leaf input was not tested.	In a brief check, images of pure random noise were returned as confident diagnoses. The model has no "this is not a leaf" class, and the thresholds did not catch them. Photographs of soil, hands or sky should be tested.

8. Recommendations

	Action	Rationale
1	Display all three possibilities in the interface	Without this the 0.35 threshold is unsafe; with it, the correct disease reaches the farmer 88% of the time
2	Test healthy leaves before release	The false-alarm rate on healthy plants is the largest unknown in this evaluation
3	Withhold rice, and treat wheat as provisional	34% first-guess accuracy on both; rice's shortlist barely exceeds chance
4	Retrain on field-realistic imagery	The model confuses crops that look nothing alike, which indicates the training images differ markedly from real photographs; this is the only route to fixing rice and wheat
5	Suppress the shortlist for potato	Three classes exist, so the shortlist lists all of them and conveys nothing

Reproduce with `venv/Scripts/python tools/test_zip.py --zip <set>.zip --send-crop`. Per-photograph predictions, confidences and shortlists for every configuration are in `test_results/`.